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1. Title of the invention: Drishti – Intelligent Surveillance and Facial Recognition System

2. Specify field of technology to which the invention belongs to:
The invention relates to the fields of Artificial Intelligence, Computer Vision, and Smart Surveillance Systems. It particularly pertains to face detection, identification, and real-time monitoring using AI-based recognition algorithms integrated with a web-based dashboard. The invention is designed for applications in public safety, educational institutions, and corporate premises, offering automated surveillance, attendance tracking, and threat identification capabilities through intelligent analytics.

3. List all the currently existing technologies/ products that are similar to your invention: -

- **CCTV Surveillance Systems** – Used for visual monitoring but rely on human observation.
- **Amazon Rekognition** – Cloud-based service for face and object recognition but dependent on internet connectivity and lacks local control.
- **Face++ (Megvii)** – Commercial facial recognition API used mainly in Asia, limited by licensing costs.
- **OpenCV + Dlib Models** – Research-grade face recognition libraries lacking integrated analytics dashboards.
- **Biometric Attendance Systems** – Provide recognition but require physical contact and have limited scalability.
- **Microsoft Azure Face API** – Cloud API providing detection and verification but unsuitable for local deployment in privacy-critical environments.

Technical details of the invention

1. Background of the invention:
In recent years, surveillance and identity management have become essential in both public and private sectors. Traditional security systems such as **CCTV networks** depend heavily on **manual monitoring**, which leads to delayed responses, human error, and limited scalability. Although modern facial recognition solutions exist, many are **cloud-dependent**, **costly**, or **privacy-invasive**, making them impractical for institutions that require **on-premise control**. "Drishti" addresses these challenges by introducing an **intelligent surveillance system** that integrates **AI-powered facial recognition (YOLOv8 + DeepFace)** with a **real-time monitoring dashboard**. Unlike conventional models that merely detect faces, Drishti performs live recognition, logging, and alerting without human intervention. The system captures real-time video input, detects and identifies faces using a pre-trained local database, and presents results visually on a dynamic dashboard. It can also log entries, track attendance, and highlight unknown or suspicious individuals. Conventional systems lack features such as **multi-face recognition**, **real-time alerts**, and **local privacy-preserving inference**. Drishti overcomes these by using optimized AI models that run on standard computing hardware, offering **accuracy, scalability, and autonomy**. The invention thus provides a **cost-effective, privacy-secure, and intelligent surveillance solution**, combining AI analytics and data visualization in one unified platform suitable for smart campuses, workplaces, and security-sensitive zones.

3. Drawbacks in existing state-of-art and how your invention helps to overcome such drawbacks: -

Existing surveillance systems either rely entirely on manual monitoring or use limited facial recognition APIs that depend on cloud computing. Such systems face several drawbacks:

- **Manual Dependency:** Continuous human observation increases the probability of oversight.
- **Cloud Dependency:** Many facial recognition APIs require an active internet connection, exposing data to privacy risks.
- **Lack of Real-Time Analysis:** Current systems fail to process live feeds and visualize analytics simultaneously.
- **No Integrated Dashboard:** Most solutions lack a single, unified platform to monitor multiple cameras, view analytics, and generate reports.

The proposed system, **Drishti**, mitigates these shortcomings through a **locally deployable AI-driven architecture**. It uses **YOLOv8** for real-time face detection and **DeepFace** for identity verification, running fully offline if needed. The system automatically logs identified individuals into a database and displays the information through an **interactive dashboard interface**. Additionally, Drishti includes a **real-time alert mechanism** for unknown faces and provides data-visualization and filtering capabilities for operators.
By merging AI detection, recognition, and visualization into one cohesive platform, the invention ensures **speed, accuracy, and security**, significantly improving surveillance efficiency.

4. The technical features of your invention which are different from the existing inventions/applications: -

The proposed invention, Drishti, exhibits several innovative features distinguishing it from conventional surveillance systems:

- **Real-Time Integration of Detection and Recognition:** Utilizes YOLOv8 for object/face detection and DeepFace for recognition, ensuring simultaneous multi-person tracking.
- **Web-Based Dashboard:** Offers live analytics, camera feeds, and alert notifications through a unified and secure web interface.
- **Local Processing:** Operates without relying on third-party cloud services, enhancing data privacy and reducing latency.
- **Modular Architecture:** Supports independent upgrades for each component—database, AI models, or interface—without affecting the rest of the system.
- **Scalable Design:** Capable of handling multiple camera inputs and expanding to large-scale institutional or city-level setups.
- **Automated Reporting:** Generates attendance logs, visitor reports, and recognition summaries for easy analysis.

These features make Drishti not only a **surveillance tool** but also an **AI-based analytical assistant** for modern infrastructure management.

5. Main advantages of your invention: -

- **Automated and Real-Time Monitoring** – Eliminates manual supervision with continuous AI-based observation.
- **Privacy-Secure Local Processing** – All recognition happens locally, avoiding cloud-based vulnerabilities.

- **Unified Dashboard Interface** – Integrates detection, recognition, and analytics in one interface.
- **Multi-Face Recognition** – Handles crowd or classroom-level analysis efficiently.
- **Instant Alerts and Logging** – Flags unknown individuals in real time.
- **Scalability and Customizability** – Easily adaptable to different scales and environments.
- **Low-Cost Implementation** – Runs efficiently on standard computing hardware without specialized GPUs.

6. Complete description of the invention –

The invention, **Drishti**, is a software-based intelligent surveillance and facial recognition system that combines advanced computer vision algorithms with a web-based monitoring interface. Its primary objective is to detect and recognize individuals in real-time through live video feeds and display the results dynamically for administrators or security personnel.

a. **Components/Embodiments involved in the invention** -

- **Camera Module:** Captures live video input and transmits frames to the backend processing unit.
- **Detection Module (YOLOv8):** Identifies human faces within frames using AI object detection.
- **Recognition Module (DeepFace):** Verifies the identity of detected faces by matching them with stored embeddings in the database.
- **Database Management System (MongoDB/MySQL):** Stores known identities, recognition logs, and attendance data.
- **Backend Server (Flask/FastAPI):** Handles data processing, API requests, and inter-module communication.
- **Dashboard Interface:** A real-time web platform displaying camera feeds, recognized individuals, time logs, and alerts for unknown entries.

b. How are the components installed/arranged? -

The system can be deployed on standard computers or local servers. The **camera module** streams input to the **backend**, which runs the detection and recognition models. Results are sent to the **database** and visualized via the **dashboard**. The user interface allows filtering, reporting, and live camera control.

The modular design allows future expansion—such as integrating emotion recognition or crowd analytics—without architectural changes.

c. Figures/Structures:

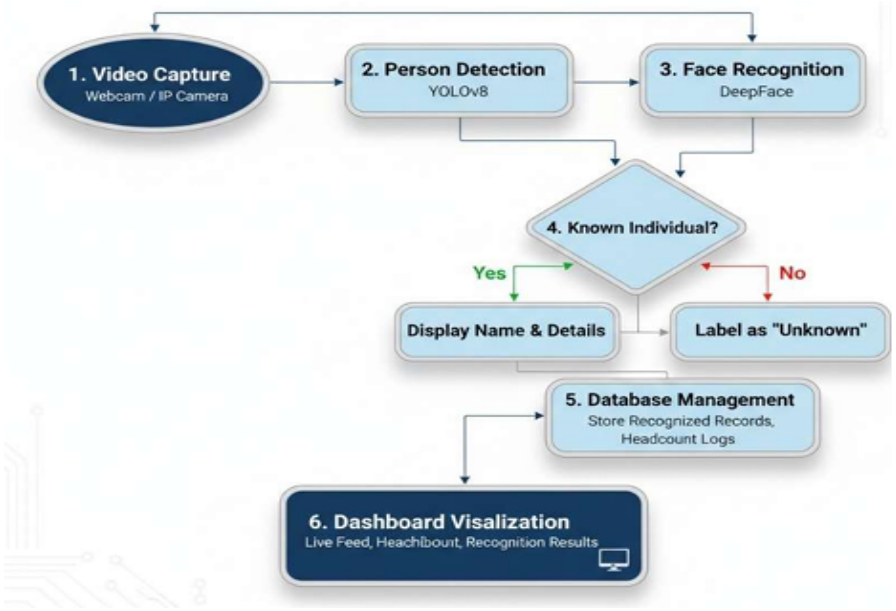

Figure 1- Operational Diagram Figure 2 -

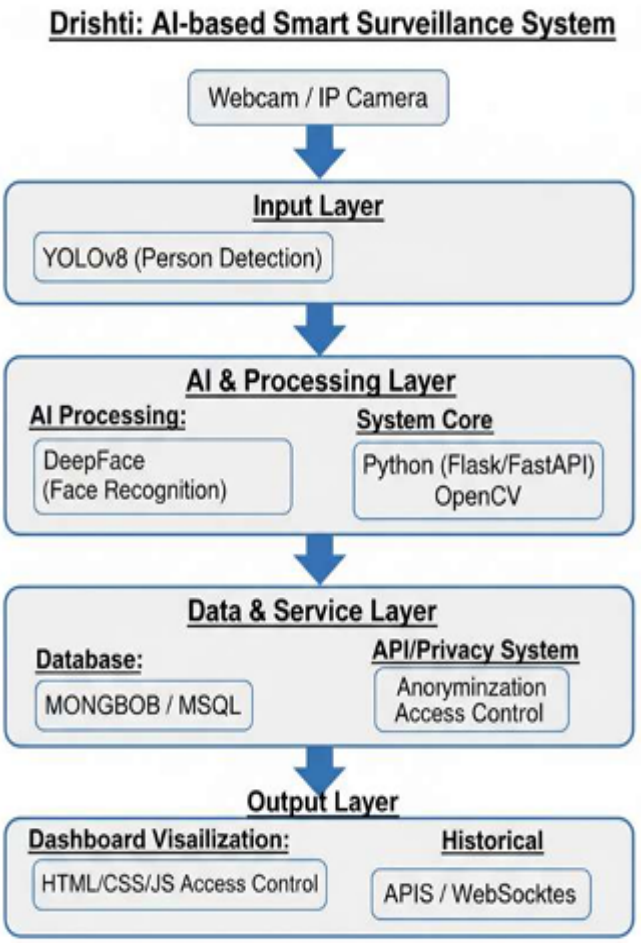

Figure 2-Detailed System Architecture

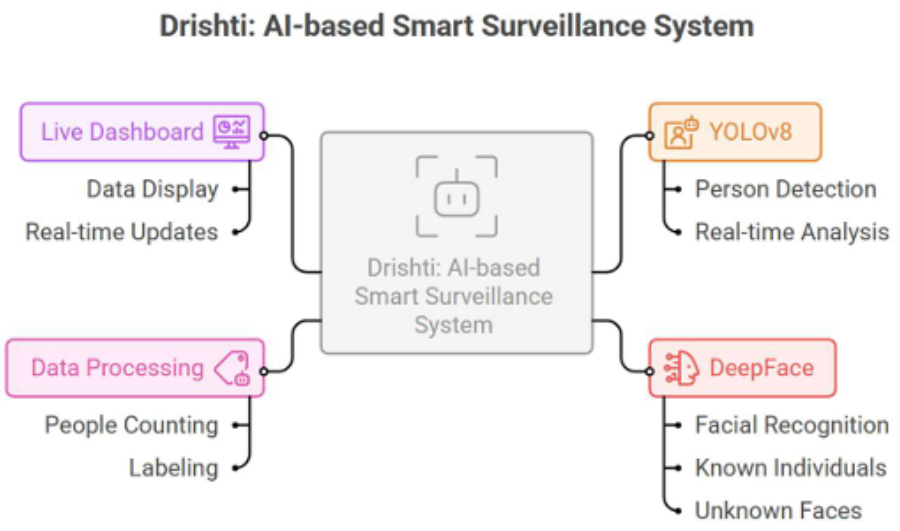

Figure 3 - System Overview

1. List of novel features in the invention -

- **Unified Dashboard-Based Surveillance System** integrating detection, recognition, and analytics.
- **Local, Privacy-Focused Face Recognition Pipeline** eliminating dependency on external cloud services.
- **Real-Time Multi-Face Recognition and Tracking** for crowd or classroom environments.
- **Alert System for Unknown Faces** to notify administrators instantly.
- **Scalable Modular Design** enabling integration with IoT cameras, access control systems, and institutional databases.
- **Offline Functionality** suitable for environments with limited connectivity.
- **Adaptable Architecture** for educational, corporate, or governmental deployment scenarios.

2. List of keywords relevant to the invention:

Face Recognition, AI Surveillance, Real-Time Detection, Intelligent Monitoring, YOLOv8, DeepFace, Dashboard Analytics, Privacy-Preserving AI, Automated Attendance, Smart Security.

3. Any other relevant details: -

- The system can be integrated into **existing CCTV infrastructures**.
- Capable of **generating attendance records automatically** in educational settings.
- Potential to extend into **gesture or emotion recognition** in future versions.
- Compliant with **privacy and data protection frameworks** when deployed locally.
- Can be adapted for **multi-language interfaces and accessibility features**.
- Future enhancements may include **voice integration**, **mobile dashboard view**, and **cloud-backup support** for hybrid architectures.